

Continuous monitoring: Continuous monitoring across all phases of the application development, testing and deployment processes is crucial for a successful DevOps implementation. The capability to improve service quality by monitoring the application's performance and log management solves the problem of aggregating, storing, and analysing all logs in one place. Splunk, ELK Stack, Nagios, Sensu and NewRelic are some of the popular tools for continuous monitoring.

Key open source tools for DevOps

Open source DevOps tools are designed and developed using open source technologies in order to provide the DevOps toolchain capabilities that are listed below:

- Planning tools that provide transparency to stakeholders and participants
 - Collaboration tools that help teams to work together, anywhere and anytime
 - Source control tools are the single source of truth
 - Issue tracking tools increase responsiveness and visibility
 - Configuration management tools enforce the desired state
 - Open source continuous integration tools work together
 - Binary repositories manage builds, releases and dependencies
 - Monitoring tools ensure service uptime and optimal performance
 - Open source automated test tools provide higher quality
 - Open source deployment tools improve time to market
- Some of the key open source DevOps tools are briefly described below.

Ansible: Red Hat owns Ansible. It automates various common tasks related to IT operations such as application deployment, configuration management and cloud provisioning. It integrates with numerous DevOps tools including Jenkins, JIRA, Git and many others. The free open source version of Ansible is available on GitHub.

Chef: Chef is an open source automation platform that transforms infrastructure into code. It operates in the cloud, on-premise or in a hybrid environment. The Chef development kit provides the tools to develop and test infrastructure automation code from a local workstation prior to deploying changes into production.

Docker: Docker is software used for OS level virtualisation. It is used to create, deploy and run application packages called containers, which allow the developer to package an application with all the parts it needs, such as libraries and other dependencies, before shipping it as one package.

Docker has two parts. Docker Engine is a tool responsible for creating and running Docker Containers. Docker Hub is a service application based on the cloud, covering the concept of application sharing and workflow automation.

GitHub: This is a collaborative code review tool supporting around 200 software languages. It also supports all the version control features of check-in, commits,

branches, merging, labels, task management, wikis, push and pull to/from GitHub, etc. Git fits in very well as a popular and distributed version control system for teams located at different geographical locations.

Many DevOps teams use it for managing the source code of their applications. It consists of plugins that can link with Jenkins to facilitate deployment and integration.

Hudson: This is a continuous integration tool developed in Java. This tool is used for managing, monitoring, continuous testing and integration. It supports various systems for the management of source code, application servers, code analysis tools, testing frameworks, build tools, real-time notifications of test failures, the changing of set support, and offers an easy process for installation and configuration.

Jenkins: Jenkins is a continuous integration tool that helps to automate the activities of build, code analysis and storing the artefacts. These activities are triggered once a developer or the team commits the code to the version control repository.

Jenkins has many plugins and works as a CI tool for various different languages like C/C++, Java/J2EE, .NET, Angular JS, etc. It also provides plugins to integrate with SonarQube for code review, JFrog Artifactory for storing binary artefacts, testing tools like Selenium, etc, as a part of the automation process.

Jenkins helps to automate deployments to app servers like Tomcat, JBoss and Weblogic through plugins and also to container platforms like Docker.

Kubernetes: Open source Kubernetes is free and downloadable from its repository on GitHub. Administrators must build and deploy the Kubernetes release to a local system or cluster; or to a system or cluster in a public cloud, such as AWS, Google Cloud Platform (GCP) or Microsoft Azure.

Puppet: This is a tool for operating and delivering software. It automates deployment to boost auditability, reliability and agility. It provides continuous automation and delivery across the complete software delivery life cycle. The latest version of Puppet features Node Manager and Puppet Apps, which help handle a large number of dynamic, variable systems.

Selenium: This is an automated, functional testing tool for Web applications. Installed as a Firefox browser plugin, it helps to record and playback test scenarios. In a DevOps scenario, once the application is deployed in a test environment, Selenium automated testing is invoked.

Supergiant: This is an open source platform for container management. It is used for Kubernetes deployment on multiple clouds in a matter of minutes. The Supergiant API is used for streamlining production deployments.

Benefits of DevOps tools

The high-level benefits of DevOps tools are:

- Reduce deployment times by 80 to 90 per cent, e.g., from 12 hours to 2 hours